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Budget Constraint - Unemployment Insurance

$$\Pi_z(u) \sum_y \Pi(y) b(y, z, \Gamma) =$$

$$\tau(z;\Gamma)[\Pi_z(u) \sum_y b(y,z,\Gamma) + (1-\Pi_z) \sum_y \Pi_y w(z,\Gamma) y$$

$\hookrightarrow \Pi_z(u) = \Pi_z$

$$\Pi_z \sum_y \Pi(y) \underbrace{b(y)}_{\rho w y} = \tau \sum_y \Pi(y)(\Pi_z \underbrace{b(y)}_{\rho w y} + (1-\Pi_z) w(z,\Gamma) y)$$

$$\Pi_z \rho w \sum_y \Pi(y) y = \tau w(\Pi_z \rho + (1-\Pi_z)) \sum_y \Pi(y) y$$

$$\tau = \Pi_z \rho / \Pi_z \rho + (1-\Pi_z) = \tau(z, \rho)$$

$\hookrightarrow$ τ does not depend on Γ since exog. unemployment variable

Social Security:
- pay tax: τ^{ss}
- pension benefit: $b^{ss}(z,\Gamma)$

$\hookrightarrow$ Budget Constraint:

$$b^{ss}(z,\Gamma) \cdot \Pi_R = \tau^{ss} \cdot \Pi_w (1-\Pi_z) \sum_y \Pi(y) w(z,\Gamma) y$$

$$b^{ss}(z,\Gamma) = \tau^{ss} \frac{\Pi_w}{\Pi_R} (1-\Pi_z) w(z,\Gamma) \underbrace{\sum_y \Pi(y) y}_{=1}$$

$$\frac{b^{ss}(z,\Gamma)}{w(z,\Gamma)} = \tau^{ss} \frac{\Pi_w}{\Pi_R}(1-\Pi_z)$$

Dynamic Programming:
↳ Retired Agents:

$$V^R(a,\beta;z,\Gamma)=\max_{c,a'}\{u(c)+\beta v\sum_{z}\Pi(z'|z)\cdot V^R(a',\beta;z',\Gamma')\}$$

s.t

$$c+a'=b^{ss}(z,\Gamma)+(1+r(z,\Gamma)-\delta)\frac{a}{v}$$

$$\Gamma'=H(\Gamma,z,z')$$

↳ $\frac{a}{v}$: accidental bequest

Dynamic Programming:
↳ Retired:

$$V^R(a,\beta;z,\Gamma)=\max_{c,a'}\{u(c)+\beta v\sum_{z}(z'|z)V^R(a',\beta;z',\Gamma')\}$$

s.t.

$$c+a'=b^{ss}(z,\Gamma)+(1+r(z,\Gamma)-\delta)\frac{a}{v}$$

$$\Gamma'=H(\Gamma,z,z')$$

$\frac{a}{v}$: bequest distributing assets of dead to living agents

Overcome accidental bequest w/ annuity market

↳ Working:

$$V^w(a,\beta,s,y;z,\Gamma)=\max_{c,a'}\{u(c)+$$

$$\beta\sum_{z'}\sum_{s'}\sum_{y'}\Pi(z'|z)\Pi(s'|s,z,z')\Pi(y'|y)$$

$$\times\ \theta V^w(a',\beta,s',y';z',\Gamma')+\beta(1-\theta)\sum_{z'}(z'|z)V^R(a',\beta;z,\Gamma')\}$$

s.t.

$$c+a'=\text{income}+(1+r(z,\Gamma)-\delta)a$$

$$\text{income}=(1-\tau(z,\rho)-\tau_c^{ss})w(z,\Gamma)y \text{ if employed}$$

$$=(1-\tau(z,\rho))\rho w(z,\Gamma)y \text{ if unemployed}$$

$$\Gamma'=H(\Gamma,Z,Z')$$

Recursive C.E.

Value and Policy Functions of Working and Retired Agents $\{V^W, V^R, a'_W, a'_R, c_W, c_R\}$

Pricing Functions w and r

Aggregate Law of Motion $H(\cdot)$

s.t.

1: Given price functions $\{w, r\}$, government policy $\{\gamma(z,\rho), b^{ss}, \rho\}$, and law of motion $\{H(\cdot)\}$ value and policy functions solve household's optimization problem

2: Factor prices satisfy Firm's FOC's

3: Budget balance for unemployment insurance and social security programs is satisfied

4: Markets clear:

$$N(z,\Gamma) = (1-\Pi_z(u))\sum_y \Pi(y)y$$
$$K(z,\Gamma) : \int a\,d\Gamma$$

5. Law of motion $H(\cdot)$ is induced by the exogenous stochastic processes for idiosyncratic and agg. risk as well as optimal policy functions.

5 is what makes this different

Homework 5 solution

1)

$\min\{c^y, \beta c^o\}$

s.t.

$K_t^\alpha L_t^{1-\alpha} + K = K_{t+1} + c_t^y L_t + c_t^o L_{t-1}$

$\hookrightarrow$ div by L_t

$k_t^\alpha + k_t = k_{t+1}(1+n) + c_t^y + \frac{c_t^o}{(1+n)}$

$\Rightarrow k^\alpha - kn = c^y + \frac{c^o}{1+n}$

$c^y = \beta c^o$

FOC:

$k: \alpha k^{\alpha-1} = n \Rightarrow k^{sp} = \left(\frac{\alpha}{n}\right)^{\frac{1}{1-\alpha}}$

$k^\alpha - k(\alpha k^{\alpha-1}) = (1-\alpha)k^\alpha = \beta c^o + \frac{c^o}{1+n}$

$c^o = \frac{(1+n)(1-\alpha)k^\alpha}{1+\beta(1+n)} = \frac{(1+n)(1-\alpha)}{(1+\beta)(1+n)}\left(\frac{\alpha}{n}\right)^{\frac{1}{1-\alpha}}$

$c^y = \beta \cdot$

2)

$\min\{c^y, \beta c^o\}$

s.t.

$c^y = w - s$

$c^o = s(1+r)$

$w - s = \beta s(1+r) \Rightarrow w = (\beta(1+r)+1)s \Rightarrow s = \frac{w}{1+\beta(1+r)}$

$c^y = \frac{w(\beta(1+r))}{1+\beta(1+r)} \Rightarrow c^o = \frac{w(1+r)}{1+\beta(1+r)}$

2)

$r = \alpha k^{\alpha-1}$

$w = (1-\alpha)k^{\alpha}$

$s = k(1+n)$ is MCC

$$\frac{w}{1+\beta(1+r)} = k(1+n) = \frac{(1-\alpha)k^{\alpha}}{1+\beta(1+\alpha k^{\alpha-1})} = k(1+n)$$

$$(1-\alpha)k^{\alpha-1} = (1+n) + \beta(1+n)(1+\alpha k^{\alpha-1})$$

$$k^{\alpha-1}(1-\alpha-\beta(1+n)\alpha) = (1+n)(1+\beta)$$

$$k^{\alpha-1} = \frac{(1+n)(1+\beta)}{1-\alpha-\beta(1+n)\alpha} = \left[\frac{1-\alpha-\beta\alpha(1+n)}{(1+n)(1+\beta)}\right]^{\frac{1}{1-\alpha}} = k^{CE}$$